Experiment 5: illumination and colour augmentation

Module of: What Tomato Counting Errors Measure

1 Protocol

1.1 Photometric perturbation

Test-time lighting change is simulated by two single-parameter interventions applied to the validation images, both in linear light rather than on the 8-bit values, so that each means what a camera would do. Exposure multiplies every channel by 2^k for $k \in [-3, 3]$ stops. White balance applies opposite gains to the red and blue channels, $1 \pm 0.35w$ for $w \in [-1.5, 1.5]$, with green fixed; positive w is warm, negative is cool. Ripeness is read along the green-to-red axis measured in Section 3.3, so a colour cast moves every fruit along the very axis the stage label encodes before the detector sees it.

Ground-truth boxes are untouched: the intervention recolours pixels and never moves a label. Because exposure shifts confidences wholesale, each level is predicted once at a confidence floor of 0.05 and then scored at every threshold in the sweep; the tables report a fixed threshold of 0.25 next to the best threshold for that level, since a fixed-threshold comparison across exposures would measure calibration rather than robustness, exactly as it does across datasets in Section 4.3.

1.2 Training-time colour augmentation

Ultralytics applies colour augmentation as an HSV jitter whose three half-widths default to $h = 0.015$, $s = 0.7$, $v = 0.4$; it is the only colour knob in the pipeline, so widening photometric augmentation in this codebase means moving those numbers. Four configurations are compared under the otherwise unchanged protocol: stock, the default recipe; hsv_v08, with brightness jitter doubled to $v = 0.8$; hsv_wide, with $v = 0.8$ and hue widened to $h = 0.05$, the nearest thing an HSV jitter has to a colour cast; and no_colour, with all three set to zero, the floor.

The test-time perturbation above is deliberately not an HSV jitter. Scoring an HSV-augmented model on HSV-jittered images would establish only that the model fitted its own augmentation; an exposure change applied in linear light is generated by a different process than the training noise.

1.3 Statistical analysis

Two sources of variation are separated. Training variation is estimated by repeating every colour-augmentation configuration at seeds 0, 1, and 2 under an identical protocol. The stock configuration is itself repeated, so the spread of a recipe against itself is the noise floor that any difference between recipes has to clear; results are reported as the mean over seeds with the observed range, and no normality is assumed from three runs.

Evaluation variation is estimated at the level of images, where the sample is larger. For a pair of models at one perturbation level, the difference in per-image counting error is resampled over the 161 validation images with 10,000 bootstrap replicates, giving a 95% confidence interval for the paired difference, and the paired null is tested by a sign-flip permutation test with 10,000 relabellings of the same per-image differences. A permutation test is preferred to a signed-rank test here because per-image counting errors are small integers with many ties, for which the rank statistic's tie correction is an approximation and the sign-flip null is exact under the paired-exchangeability hypothesis. Detection recall is a ratio of pooled counts rather than a per-image mean, so its interval is obtained from the same bootstrap applied to the per-image matched and ground-truth counts. Where a family of levels is tested together, p-values are adjusted across that family with the Holm correction.

Per-image series are averaged over seeds before the paired test, so the reported interval describes the precision of this validation set for a recipe rather than for one checkpoint.

Both estimates are reported, because they answer different questions: the bootstrap interval states how precisely this validation set measures a given pair of checkpoints, while the seed range states how much of the difference would survive retraining.

2 Results

2.1 Lighting moves the detector more than capacity does

A change of light is a larger perturbation than a change of model. Table 1 places both on the same scale: moving from YOLO11n to YOLO11x buys 0.058 recall on clean frames, while over-exposing those same frames by three stops costs the best model on the ladder 0.205. Every size loses more to light than the whole ladder spans.

Under-exposure is nearly free and over-exposure is not. Down to -3 stops — a frame with mean luma 42 out of 255 — recall falls by at most 0.03 for any model, and for the widest-jitter variants it does not fall at all. At $+3$ stops recall drops by 0.15 to 0.21 and counting MAE roughly doubles. The asymmetry has a physical cause visible in Figure 1: darkening compresses the signal but preserves the ratios between channels, whereas clipping destroys them, and a blown-out red tomato has lost the saturation the ripeness

Extract built by python paper/build.py lighting. Section, table and figure numbers are those of the full paper.



Fig. 1. One validation frame under the photometric interventions: exposure from −3 to +3 stops (top) and white balance from cool to warm (bottom). Captions give the mean luma of the frame and the baseline detector's recall over the whole validation split at that level. Annotations are never moved; only the pixels change.

TABLE 1
Recall at confidence 0.25 and counting MAE at the best threshold under the four extreme lighting conditions. One seed per row; the colour-augmentation rows are repeated over three seeds in Table 2.

Model	clean	−3 st.	+3 st.	cool	warm
YOLO11n	0.877	0.860	0.696	0.797	0.817
no colour jitter	0.877	0.765	0.608	0.751	0.782
hsv_v08	0.874	0.870	0.692	0.810	0.820
hsv_wide	0.871	0.868	0.704	0.824	0.847
YOLO11s	0.916	0.904	0.739	0.842	0.883
YOLO11m	0.935	0.901	0.786	0.870	0.898
YOLO11l	0.929	0.912	0.782	0.877	0.901
YOLO11x	0.935	0.914	0.730	0.865	0.899
counting MAE at the best threshold					
YOLO11n	0.96	1.13	1.78	1.31	1.35
no colour jitter	1.00	1.47	2.07	1.52	1.71
YOLO11m	0.88	1.05	1.68	1.11	1.27
YOLO11x	0.87	0.93	1.81	1.23	1.06

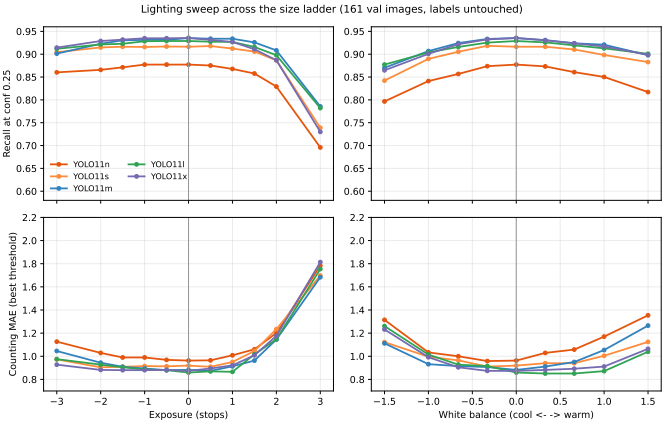


Fig. 2. Recall at a fixed threshold and counting error at the best threshold along the exposure and white-balance axes for the whole size ladder. The curves are ordered by capacity on clean frames and converge at the extremes.

stage is read from. Capacity does not repair that: YOLO11x is no better at +3 stops than YOLO11s.

The two white-balance directions are not symmetric either. A cool cast costs more recall than a warm one at the same magnitude, while a warm cast costs counting more than it costs detection, because it pushes green and half-ripened fruit along the green-to-red axis toward riper calls and the model over-counts the ripe stage. This is the same mechanism that Section 3.3 measures between two annotation conventions, arriving this time through the camera rather than through the annotator.

2.2 Does wider colour augmentation help?

The supervising question behind this section was whether photometric augmentation should be strengthened to simulate lighting conditions. Answering it needs the noise floor first, because the effects involved are of the same order as the spread between retrainings.

TABLE 2
Colour-augmentation recipes over three seeds: recall at confidence 0.25, mean (observed range). The spread of the default recipe against itself is the noise floor for this comparison.

Recipe	clean	−3 st.	+3 st.	cool
default	0.875	0.859	0.666	0.792
	(0.874–0.877)	(0.857–0.860)	(0.647–0.696)	(0.773–0.807)
hsv_v08	0.872	0.872	0.707	0.794
	(0.868–0.875)	(0.870–0.874)	(0.692–0.728)	(0.774–0.810)
hsv_wide	0.874	0.870	0.694	0.808
	(0.871–0.880)	(0.862–0.881)	(0.686–0.704)	(0.781–0.824)
no_colour	0.876	0.757	0.604	0.758
	(0.872–0.879)	(0.739–0.768)	(0.567–0.636)	(0.731–0.791)

That spread is not constant. Retraining the same default recipe at three seeds changes clean recall by 0.003, but changes recall at +3 stops by 0.049, from 0.647 to 0.696 (Table 2). The harsher the perturbation, the less a single training run says

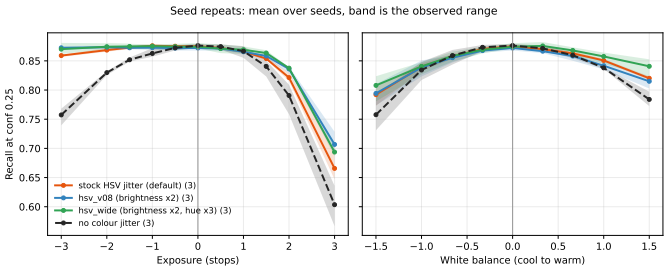


Fig. 3. Three seeds per training recipe. The line is the mean over seeds and the band is the observed range. Removing colour jitter separates from the three jittered recipes only once the frame is perturbed, and the seed spread itself widens with the perturbation.

TABLE 3
Each recipe against the default, seed-averaged and paired over the 161 validation images. ΔMAE is the change in counting error, negative favouring the recipe; Δrecall is at confidence 0.25. Brackets are 95% bootstrap intervals and p is Holm-adjusted over the five levels.

Recipe	Level	ΔMAE [95% CI]	Δrecall
hsv_v08	clean	+0.005 [−0.039, +0.049]	−0.003
	−3 st.	−0.075 [−0.130, −0.022]*	+0.013
	+3 st.	−0.136 [−0.260, −0.039]*	+0.041
	cool	−0.062 [−0.133, +0.008]	+0.002
	warm	+0.039 [−0.021, +0.101]	−0.005
hsv_wide	clean	+0.062 [+0.007, +0.116]	−0.001
	−3 st.	−0.035 [−0.101, +0.029]	+0.011
	+3 st.	−0.045 [−0.138, +0.039]	+0.028
	cool	−0.168 [−0.261, −0.075]**	+0.016
	warm	−0.001 [−0.108, +0.093]	+0.021
no_colour	clean	−0.043 [−0.103, +0.017]	+0.001
	−3 st.	+0.324 [+0.208, +0.450]***	−0.102
	+3 st.	+0.165 [+0.058, +0.280]**	−0.062
	cool	+0.203 [+0.115, +0.295]***	−0.034
	warm	+0.226 [+0.100, +0.364]**	−0.036

*$p < 0.05$, **$p < 0.01$, ***$p < 0.001$, Holm-adjusted

about the recipe — and the single-seed figure reported for the default at +3 stops in Table 1, 0.696, turns out to be the most favourable of its three runs. Any claim about colour augmentation that rests on one run at an extreme level is inside this spread.

With that established, the two findings separate cleanly.

Removing colour jitter is expensive, and invisible in-distribution. On clean frames no_colour is indistinguishable from the default: +0.001 recall, and a counting error 0.043 lower with an interval that spans zero ($p = 0.16$). Under every perturbation it loses decisively — −0.102 recall at −3 stops, −0.062 at +3, −0.034 and −0.036 under the two colour casts, with counting error worse by 0.17 to 0.32 and Holm-adjusted $p \leq 0.006$ at every one of the four levels (Table 3). An ablation scored only on held-out clean frames would have recommended deleting the augmentation that supplies all of this.

Widening the jitter helps, but only along its own axis, and by an order of magnitude less. Doubling the brightness half-width improves the exposure extremes and nothing else: −0.136 counting error and +0.041 recall at +3 stops, −0.075 and +0.013 at −3 stops, both significant after correction, while neither white-balance level separates from the default. Widening hue as well moves the other axis instead: −0.168 counting error at a cool cast ($p = 0.002$) with +0.016 recall, and +0.021 recall at a warm cast, while its exposure gains fall inside the interval and it pays a small penalty on clean frames (+0.062 counting error, not significant after correction).

The mechanism is therefore legible rather than mysterious: an augmentation transfers to the perturbation it resembles. A brightness jitter is an exposure simulator and buys exposure robustness; a hue jitter is the closest an HSV pipeline comes to a colour cast and buys white-balance robustness. What neither buys is general photometric robustness.

The sizes matter for the recommendation. Removing the default jitter costs up to 0.102 recall; widening it returns at most 0.041, and only where it matches the perturbation. The default recipe is therefore doing most of the available work already, and the practical advice is to keep it rather than to tune it — while noting that if a deployment is known to face one axis, for instance harsh backlight, widening that specific half-width is a real if modest gain.

These conclusions rest on three seeds per recipe and on paired tests over 161 images. The paired intervals describe this validation set precisely; the seed ranges show that the smaller effects, those under about 0.02 recall, would not be reliably reproduced by a single retraining.